Scenario UAT-01: Configuration and Currency Analysis Generation

Objective: Verify that currency parameter configuration works correctly and that client-side request chunking successfully merges data for periods exceeding 93 days.

Preconditions: The user has accessed the application via a web browser with an active internet connection.

Steps:

1. Select a Base Currency from the top navigation dropdown.
2. Select a Quote Currency from the adjacent dropdown.
3. Select a predefined historical timeframe that exceeds 93 days (e.g., 1 Year) from the Timeframe Control.
4. Click the **Generate Analysis** button.

Expected Results:

- The system validates the input parameters and initiates data fetching from the NBP Web API.
- The system splits the 1-year request into sequential chunks (max 93 days each), executes them asynchronously, and merges the responses into a single dataset.
- Visual feedback (skeleton loaders or pulse animations) is displayed during API loading times to prevent UI freezes.
- The application state updates seamlessly with the consolidated dataset.

Scenario UAT-02: Statistical Measures & Session Trends Evaluation

Objective: Verify that the client-side engine correctly evaluates day-to-day changes and outputs precise statistical metrics.

Preconditions: Scenario UAT-01 has been executed successfully, and the data is fully loaded into the application state.

Steps:

1. Observe the "Session Summary" panel on the dashboard.
2. Observe the "Statistical Measures" grid on the dashboard.

Expected Results:

- The client-side statistical engine processes the aggregated data completely within the browser without any backend server dependencies.
- The system displays the correct counts for upward (growth), downward (decline), and unchanged (stable) sessions.
- The grid displays the correct calculated values for Median, Mode, Standard Deviation, and Coefficient of Variation.
- No NaN or Infinity values are present; metrics are properly formatted.

Scenario UAT-03: Change Distribution Analysis and Timeframe Toggling

Objective: Verify that the system correctly calculates session-to-session differences and updates the histogram dynamically based on custom timeframes and aggregation toggles.

Preconditions: Scenario UAT-01 has been executed successfully.

Steps:

1. Navigate to the "Change Distribution Analysis" section at the bottom of the dashboard.
2. Click the aggregation mode toggle to switch between **Month** and **Quarter**.
3. Use the calendar Start Date Picker to select a custom past date in history.

Expected Results:

- The system accurately calculates exchange rate differences between consecutive sessions for the designated period.
- The calculated differences are grouped into mathematical frequency ranges.
- A Recharts-powered frequency histogram is rendered on-screen, showing the shape of market volatility.
- The adjacent Frequency Results Table displays matching, precise numerical session counts for each range interval.

Scenario UAT-04: Exporting Analysis Results and Visualizations

Objective: Verify that the client-side data parsing can successfully trigger file downloads for CSV, JSON, and image formats.

Preconditions: Data has been successfully fetched and evaluated; the analytical dashboard is completely rendered.

Steps:

1. Click the **Export Data** button in the bottom-right corner of the interface.
2. Select **CSV** from the available format options and save the file.
3. Re-open the export option, select **JSON**, and save the file.
4. Select **PNG/JPEG** to export the visual chart.

Expected Results:

- The system dynamically parses the raw data, calculated statistics, and frequency bins into the selected format.
- For image exports, the system converts the Recharts SVG elements into a high-resolution image using a canvas-based utility.
- A temporary Blob URL/Data URL is created, and the browser automatically triggers a file download to the local device.

Scenario UAT-05: Alternative Flows and Error Handling

Objective: Verify the resilience of the application when encountering NBP server errors or empty data sets.

Preconditions: The application is open in the browser.

Steps (Test A - API Outage):

1. Simulate an unreachable NBP API or server timeout, then click **Generate Analysis**.

Steps (Test B - Non-Trading Windows):

1. Select a custom historical timeframe that consists entirely of non-trading days (e.g., a single holiday weekend) and trigger the analysis.

Expected Results:

- **Test A Result**: If the NBP API fails to respond, the system does not crash; it catches the error and displays a user-friendly message advising the user to try again later.
- **Test B Result**: The system alerts the user with a notification stating that no data is available for that specific non-trading period.